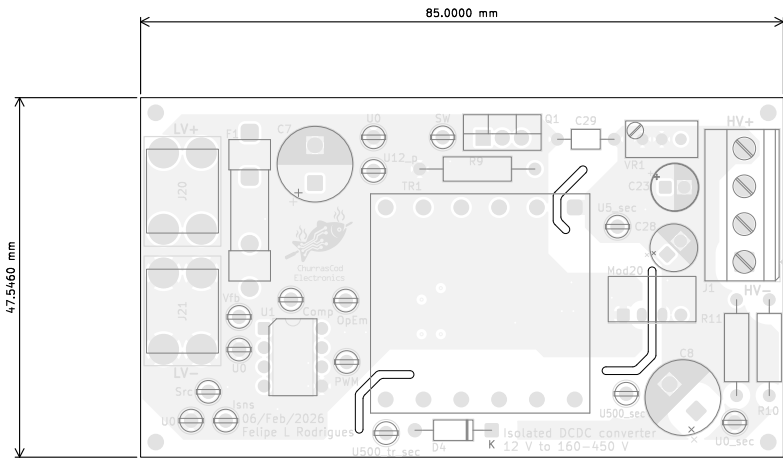


Step-Up Fabrication Document

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

Layer Stack Legend

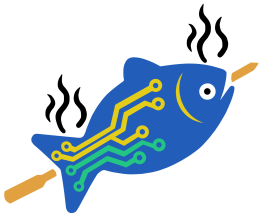
| Material     | Layer                      | Thickness        | Dielectric      | Type         | Gerber |
|--------------|----------------------------|------------------|-----------------|--------------|--------|
| F.Paste      |                            |                  |                 | Paste Mask   |        |
| F.Silkscreen |                            |                  | Direct Printing | Legend       | GBR    |
| F.Mask       |                            | 0.02mm           | Solder Resist   | Solder Mask  | GBR    |
| Copper       | L1 (Ref)                   | 0.07mm (2.00oz)  |                 | Plane        | GBR    |
| Prepreg      |                            | 0.001mm          | FR4_7628        | Dielectric   |        |
| Copper       | Virtual (Top side Jumpers) | 0.035mm (1.00oz) |                 | Copper Layer | GBR    |
| Core         |                            | 1.35mm           | FR4             | Dielectric   |        |
| Copper       | Virtual (Bot side Jumpers) | 0.035mm (1.00oz) |                 | Copper Layer | GBR    |
| Prepreg      |                            | 0.001mm          | FR4             | Dielectric   |        |
| Copper       | L2 (Sig, PWR)              | 0.07mm (2.00oz)  |                 | Signal       | GBR    |
| B.Mask       |                            | 0.02mm           | Solder Resist   | Solder Mask  | GBR    |
| B.Silkscreen |                            |                  | Direct Printing | Legend       | GBR    |
| B.Paste      |                            |                  |                 | Paste Mask   |        |

Total thickness: 1.602mm  
Note: external layer thicknesses are specified after plating

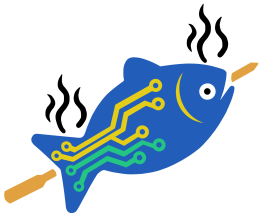
FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

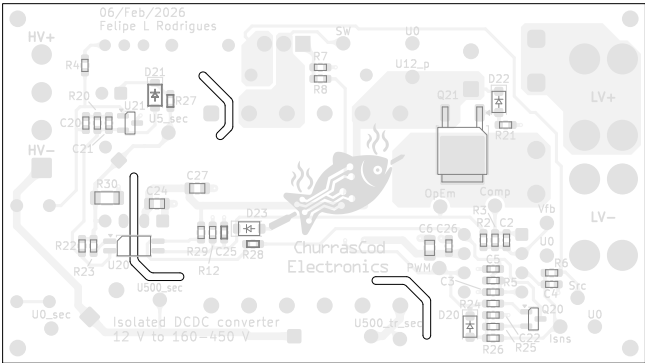
- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge\_Cuts.GBR" SUFFIX.  
  
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.  
  
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SURFACE FINISH: IMMERSION GOLD
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:  
  
A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.  
B. Tg 170 C OR EQUIVALENT.  
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY NA.
- DESIGN GEOMETRY MINIMUM FEATURE SIZES:

|                  |                    |
|------------------|--------------------|
| BOARD SIZE       | 85,000 × 47,546 mm |
| BOARD THICKNESS  | 1.602 mm           |
| TRACE WIDTH      | 0.400 mm           |
| TRACE TO TRACE   | -0.000 mm          |
| MIN. HOLE (PTH)  | 0.250 mm           |
| MIN. HOLE (NPTH) | 2.100 mm           |
| ANNULAR RING     | 0.150 mm           |
| COPPER TO HOLE   | 0.200 mm           |
| COPPER TO EDGE   | 0.250 mm           |
| HOLE TO HOLE     | 0.254 mm           |

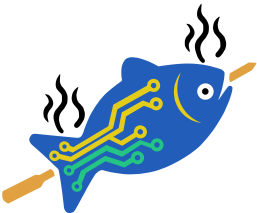


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|                                                                                      |                                             |                                           |                 |                                            |                      |
|--------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------|-----------------|--------------------------------------------|----------------------|
|  | Comments:                                   | Company:<br>NA                            |                 | Variant:<br>RELEASED                       | Git Hash:<br>fb82674 |
|                                                                                      | Sheet Title:<br>Top Fabrication (Scale 1:1) | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                      |
|                                                                                      | Sheet Path:                                 | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0   |
|                                                                                      |                                             |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>1 of 7     |



All dimensions are in millimeters unless otherwise specified.



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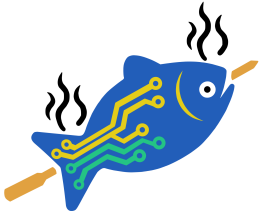
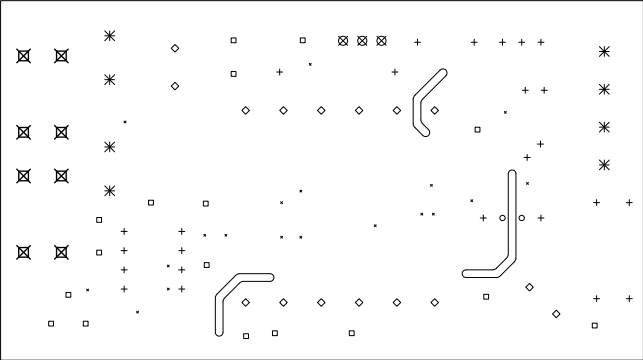
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| Sheet Title:<br>Bottom Fabrication (Scale 1:1) | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0   |
| Sheet Path:                                    |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>2 of 7     |

# Step-Up Fabrication Document

## Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

| Symbol   | Count | Hole Size          | Plated | Hole Shape | Drill Layer Pair         | Hole Type |
|----------|-------|--------------------|--------|------------|--------------------------|-----------|
| ×        | 19    | 0,25mm (9,84mils)  | PTH    | Round      | L1 (Ref) - L2 (Sig, PWR) | Via       |
| ○        | 2     | 0,70mm (27,56mils) | PTH    | Round      | L1 (Ref) - L2 (Sig, PWR) | Pad       |
| +        | 25    | 0,80mm (31,50mils) | PTH    | Round      | L1 (Ref) - L2 (Sig, PWR) | Pad       |
| □        | 17    | 0,90mm (35,43mils) | PTH    | Round      | L1 (Ref) - L2 (Sig, PWR) | Pad       |
| ◇        | 16    | 1,00mm (39,37mils) | PTH    | Round      | L1 (Ref) - L2 (Sig, PWR) | Pad       |
| ⊠        | 3     | 1,20mm (47,24mils) | PTH    | Round      | L1 (Ref) - L2 (Sig, PWR) | Pad       |
| ✱        | 8     | 1,30mm (51,18mils) | PTH    | Round      | L1 (Ref) - L2 (Sig, PWR) | Pad       |
| ⊠        | 8     | 1,80mm (70,87mils) | PTH    | Round      | L1 (Ref) - L2 (Sig, PWR) | Pad       |
| Total 98 |       |                    |        |            |                          |           |



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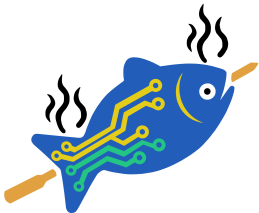
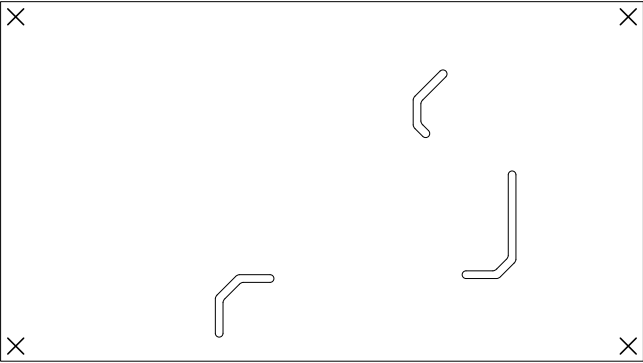
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| Comments:                               | Company:<br>NA                            |                 | Variant:<br>RELEASED                       | Git Hash:<br>fb82674 |
|                                         | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                      |
| Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0   |
| Sheet Path:                             |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>3 of 7     |

# Step-Up Fabrication Document

## Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

| Symbol | Count   | Hole Size          | Plated | Hole Shape | Drill Layer Pair         | Hole Type  |
|--------|---------|--------------------|--------|------------|--------------------------|------------|
| X      | 4       | 2.10mm (82.68mils) | NPTH   | Round      | L1 (Ref) - L2 (Sig, PWR) | Mechanical |
|        | Total 4 |                    |        |            |                          |            |



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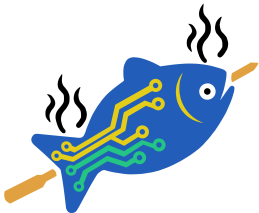
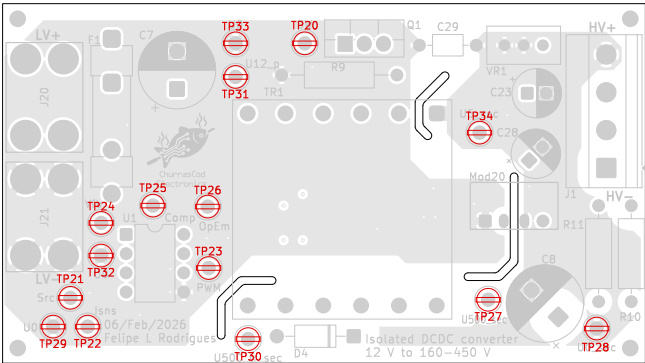
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|                                         | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                      |
| Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0   |
| Sheet Path:                             |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>4 of 7     |

Step-Up Fabrication Document

Top Test Points (Scale 1:1)

| Ref. | Net         | X [mm] | Y [mm] |
|------|-------------|--------|--------|
| TP20 | SW          | 41.95  | 44.73  |
| TP21 | Src         | 10.97  | 11.07  |
| TP22 | Isns        | 13.25  | 7.26   |
| TP23 | PWM         | 29.25  | 15.01  |
| TP24 | Vfb         | 15.03  | 20.98  |
| TP25 | Comp        | 21.89  | 23.27  |
| TP26 | OpEm        | 29.13  | 23.14  |
| TP27 | U500_sec    | 66.21  | 10.82  |
| TP28 | U0_sec      | 80.56  | 7.01   |
| TP29 | U0          | 8.68   | 7.26   |
| TP30 | U500_tr_sec | 34.46  | 5.61   |
| TP31 | U12_p       | 32.81  | 40.28  |
| TP32 | U0          | 15.03  | 16.66  |
| TP33 | U0          | 32.81  | 44.73  |
| TP34 | U5_sec      | 65.07  | 32.92  |

All dimensions are in millimeters unless otherwise specified.

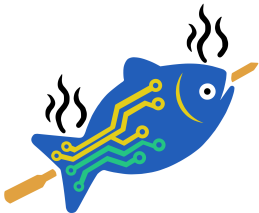
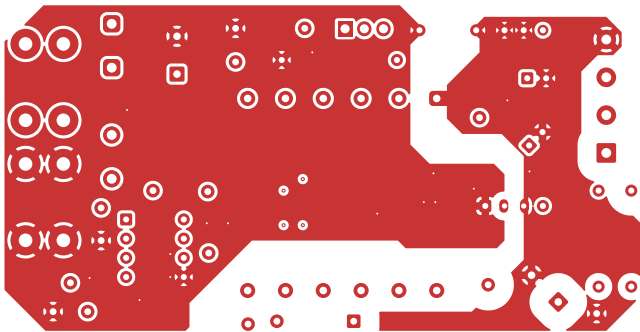


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|                                             |                                           |  |                      |                                            |                      |  |                    |  |
|---------------------------------------------|-------------------------------------------|--|----------------------|--------------------------------------------|----------------------|--|--------------------|--|
| Comments:                                   | Company:<br>NA                            |  | Variant:<br>RELEASED |                                            | Git Hash:<br>fb82674 |  |                    |  |
|                                             | Board Name:<br>Step-Up                    |  |                      | Project Name:<br>Isolated step-up 12:450 V |                      |  |                    |  |
| Sheet Title:<br>Top Test Points (Scale 1:1) | File Name:<br>Isolated_12to450V.kicad_pcb |  | Designer:<br>FR      |                                            | Date:<br>2026-02-07  |  | Revision:<br>2.0.0 |  |
| Sheet Path:                                 |                                           |  | Reviewer:<br>FR      |                                            | Size:<br>A4          |  | Sheet:<br>5 of 7   |  |

Step-Up Fabrication Document

F.Cu (Scale 1:1)

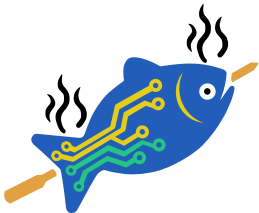
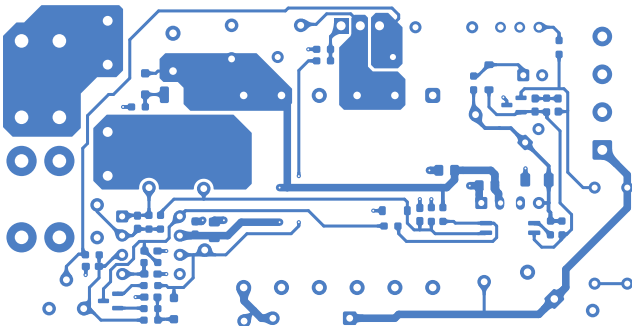


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|              |                             |           |                           |           |
|--------------|-----------------------------|-----------|---------------------------|-----------|
| Comments:    | Company:                    |           | Variant:                  | Git Hash: |
|              | NA                          |           | RELEASED                  | fb82674   |
|              | Board Name:                 |           | Project Name:             |           |
| Sheet Title: | Step-Up                     |           | Isolated step-up 12:450 V |           |
|              | F.Cu (Scale 1:1)            |           | Date:                     | Revision: |
|              | Isolated_12to450V.kicad_pcb |           | 2026-02-07                | 2.0.0     |
| Sheet Path:  |                             | Designer: | Size:                     | Sheet:    |
|              |                             | FR        | A4                        | 6 of 7    |
|              |                             | Reviewer: |                           |           |
|              |                             | FR        |                           |           |

Step-Up Fabrication Document

B.Cu (Scale 1:1)



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Electronics

|                                  |                                           |                 |                                            |                      |
|----------------------------------|-------------------------------------------|-----------------|--------------------------------------------|----------------------|
| Comments:                        | Company:<br>NA                            |                 | Variant:<br>RELEASED                       | Git Hash:<br>fb82674 |
|                                  | Board Name:<br>Step-Up                    |                 | Project Name:<br>Isolated step-up 12:450 V |                      |
| Sheet Title:<br>B.Cu (Scale 1:1) | File Name:<br>Isolated_12to450V.kicad_pcb | Designer:<br>FR | Date:<br>2026-02-07                        | Revision:<br>2.0.0   |
| Sheet Path:                      |                                           | Reviewer:<br>FR | Size:<br>A4                                | Sheet:<br>7 of 7     |